Tallet 2 Fisica

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2

$vc = 20.0 \frac{m}{s}$

$a = 3.20 \frac{m}{s^2}$

$vo = 0$

$d = \cancel{vot} + \frac{1}{2} at^2 + \cancel{do}$

$aa = \frac{vf - vo}{t}$

a) $x = 20.0 \frac{m}{\cancel{s}} \times 12.5 \cancel{s}$

$= 250.0 m$

$\boxed{vf = vo + at}$

$vA = 46.0 \frac{m}{s}$

camión: $x = vt$

$x = 20.0 \frac{m}{s} \cdot t$

Auto $= x = \frac{1}{2} at^2$

$x = \frac{1}{2} \times 3.20 \frac{m}{s^2} \cdot t^2$

$x = 1.60 \frac{m}{s^2} \cdot t^2$

$20.0 \frac{m}{s} t = 1.60 \frac{m}{s^2} \cdot t^2$

$20.0 = \frac{1.60 t}{s}$

$20.0 s = 1.60 t$

$t = \frac{20.0 s}{1.60}$ $t = 12.5 s$

B)

$a = \frac{vf - vo}{t}$

$vf = vo + at$

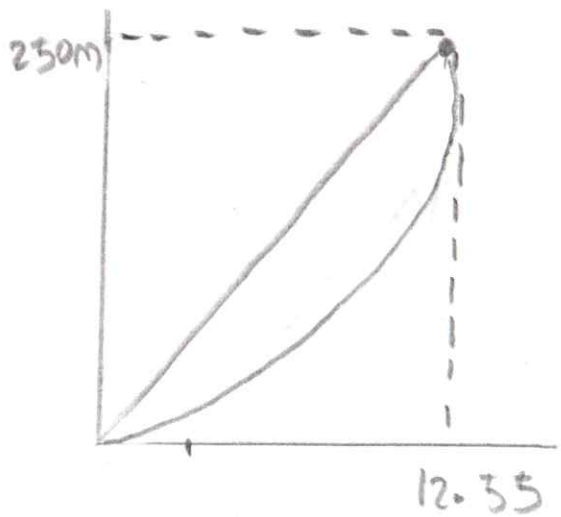
$vf = at = 3.20 \frac{m}{s^2} \times 12.5s$

$vf = 40.0 \frac{m}{s}$

c)

Camión : $x = vct$

$x = 20t$

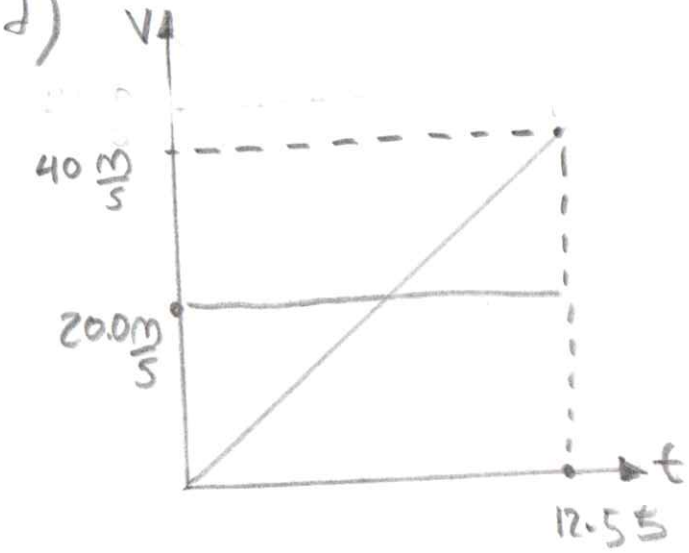


Auto : $x = vot + \frac{1}{2}at^2 + xo$

$x = \frac{1}{2} \times 3.20 \frac{m}{s^2} \cdot t^2$

$x = 1.60t^2$

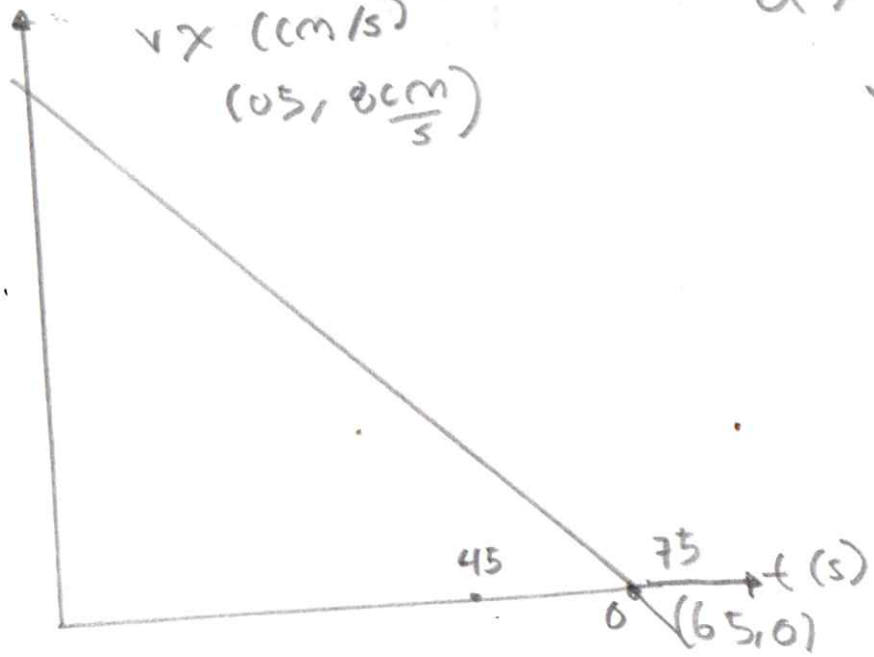
d)



$V_F = V_0 + at$

$V(t) = 3.20t$

3)



a) $v(4s) = ?$

$v(7s) =$

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$a = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\frac{8cm}{s} - \frac{0cm}{s}}{0s - 6s}$

$a = \frac{8cm/s}{-6s} = -1.33 \frac{cm}{s^2}$

$$x = \left(\frac{8}{s}t - \frac{1}{2}\frac{1.333}{s^2}t^2\right)$$

$$Vf = vo + at$$

$$X = 8t - 0.666t^2$$

a) $v(4s) = 8 - 1.333(4)$

$= 2.68 \frac{cm}{s}$

$V = 8 - 1.333t$

$V = 8 - 1.333t$

$v(7) = 8 - 1.33(7) = -1.32 \frac{cm}{s}$

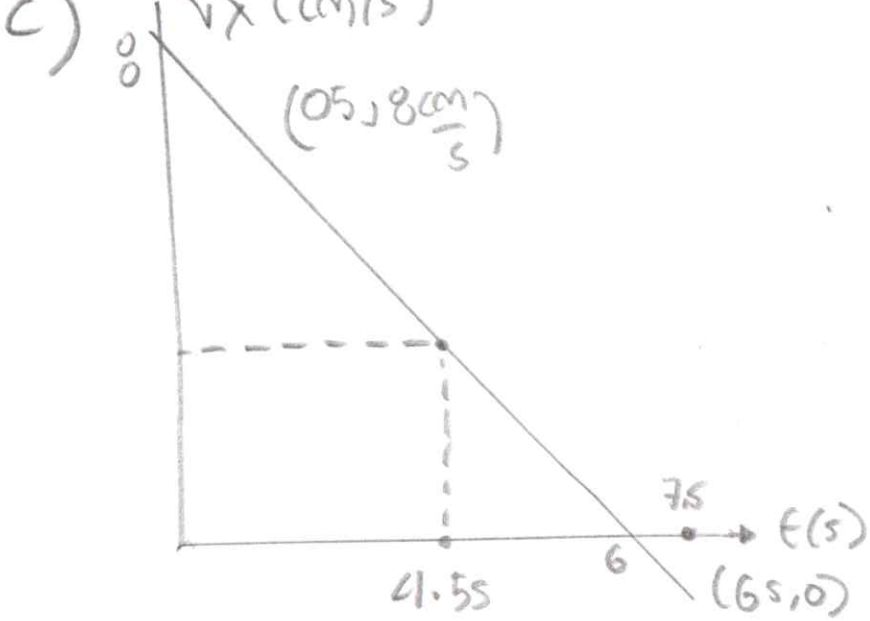
la aceleracion permanece constante en el tiempo $= -1.33 \frac{cm}{s^2}$

b) $a = \frac{vf - vo}{t}$

$V(3) = 8 - 1.333(3) = 4.01 \frac{cm}{s}$

$a = \frac{4.01 - 8}{3} = -1.33 \frac{cm}{s^2}$

c)



$$V_F = 8 - 1.33(4.5)$$

$$V_F = 2.015 \frac{cm}{s^2}$$

$$\left(\frac{B+b}{2}\right)h = \text{Area trapecio}$$

$$x = \left(\frac{8 + 2.015}{2}\right) \cdot 4.5$$

$$x = 22.53 \text{ cm}$$

$$V_F = V_0 + at$$

$$V(t) = 8 - 1.33t$$

$$\frac{dx}{dt} = 8 - 1.33t$$

$$dx = (8 - 1.333t)dt$$

$$\int_{x_0}^{x_F} dx = \int_0^{t_F} (8 - 1.333t))\, dt$$

$$x \Big|_{x_0}^{x_F} = 8t - \frac{1.333t^2}{2} \Big|_0^{t_F}$$

$$x_F - x_0 = 8t_F - 0.667t_F^2$$

$$x(t) = 8t - 0.667t^2$$

$$x(4.5) = 8(4.5) - 0.667(4.5)^2$$

$$x = 22.49 \text{ cm}$$

$$\frac{\quad}{2} \qquad \frac{\quad}{2}$$

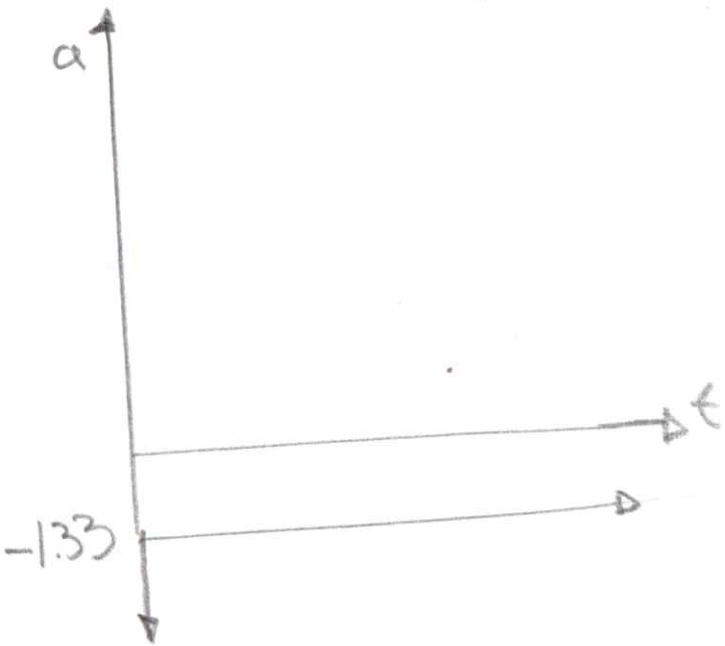
$v_f = 8 - 1.33(7.5)$ $\qquad = 25.48\,cm$

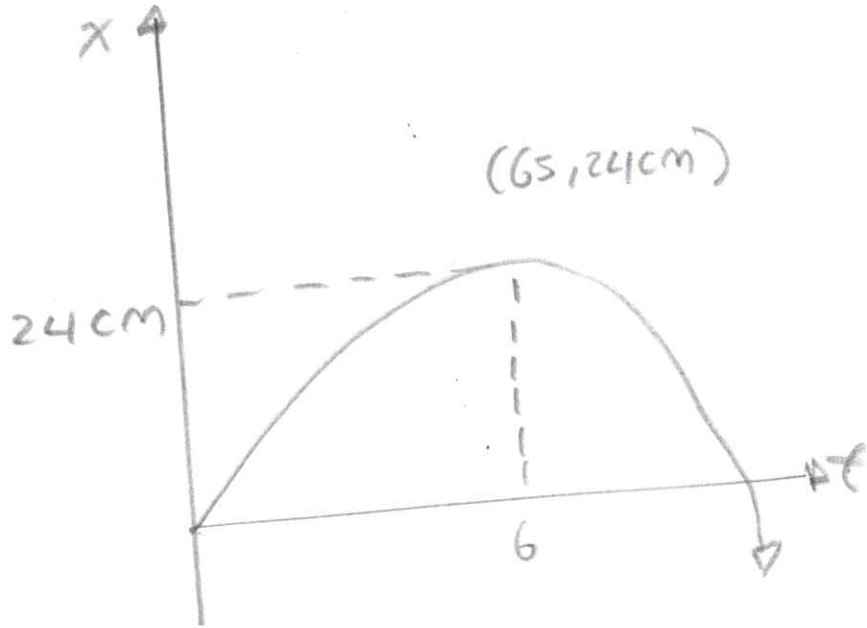
$= -1.975 \frac{cm}{s}$

D) $x(t) = 8t - 0.667t^2$

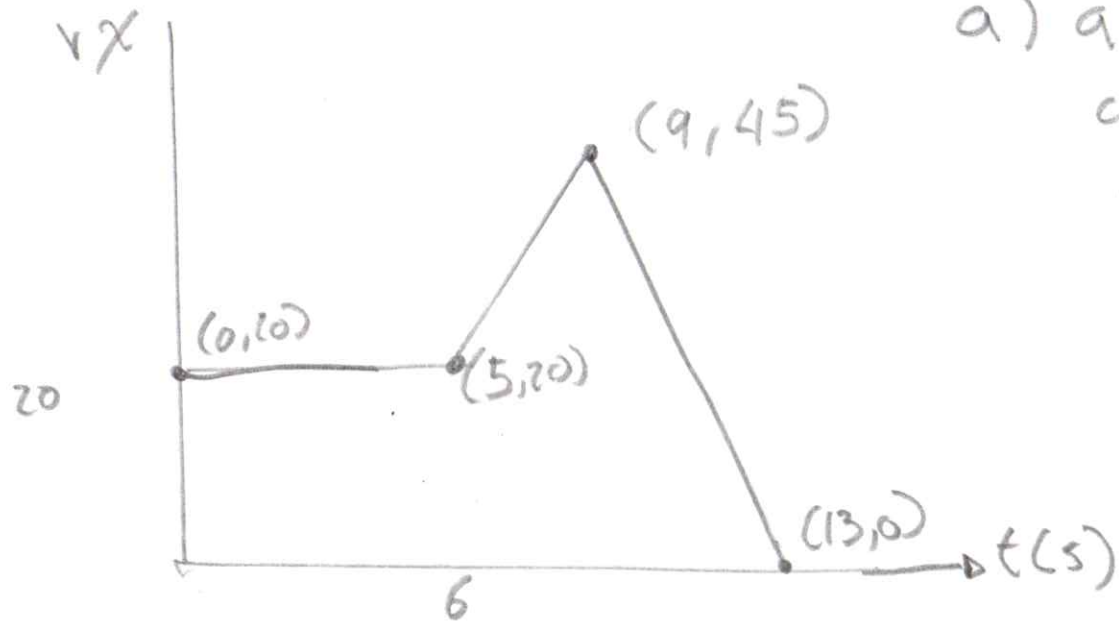
$v(t) = 8 - 1.33t$

$a(t) = -1.33 \frac{cm}{s^2}$





4)



a) $a(3s) = ?$

$a(7s) = ?$

$a(11s) = ?$

$m = \frac{\Delta y}{\Delta x}$

$a = \frac{\Delta V}{\Delta t} = \frac{V_2 - V_1}{t_2 - t_1} =$

$\frac{20-20}{3-0} = \frac{0}{3} = 0$

$a(3s) = 0$

$a(7s) = \frac{45-20}{9-5}$

$a(7s) = 6.25 \frac{m}{s^2}$

$a(11s) = \frac{45-0}{9-13} = \frac{45}{-4} = -11.25 \frac{m}{s^2}$

b) $x(5s) = 5\cancel{s} \times \frac{20m}{\cancel{s}} = 100m$

$x(9s) = 100m + \frac{\left(\frac{45m}{s} + 20m/s\right)}{2} \times 4s$

$x(9s) = 100m + 130m = 230m$

$x(13s) = 230m + 4s \times \frac{\frac{45m}{s}}{2}$

$x(13s) = 320m$

5)

$$\text{Rapidez} = \frac{\pi r}{10s} = \frac{3.1416 \times 25cm}{10s} = 78.54 \frac{cm}{s}$$

$$\text{velocidad} = \frac{\Delta x}{\Delta t} = \frac{x2 - x1}{10s} = 7.85 \frac{cm}{s}$$

$$\frac{\text{Diametro}}{\text{tiempo}}$$

$$\frac{\frac{1}{2}\pi r}{10s} \rightarrow \frac{50cm}{10s} = \frac{5cm}{s}$$

$s = 2\pi r \rightarrow$ ○

○ $s = \pi r$

diametro = 50.0cm

$r = 25cm$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{10 \pm \sqrt{100 - 4(0.05)(200)}}{2(0.05)}$$

$$t_1 = \frac{10 \pm 7.746}{0.10} = 177.46s \qquad t_2 = \frac{10 - 7.746}{0.10} = 22.54s$$

Sí chocan

lo alcanza en 22.54s

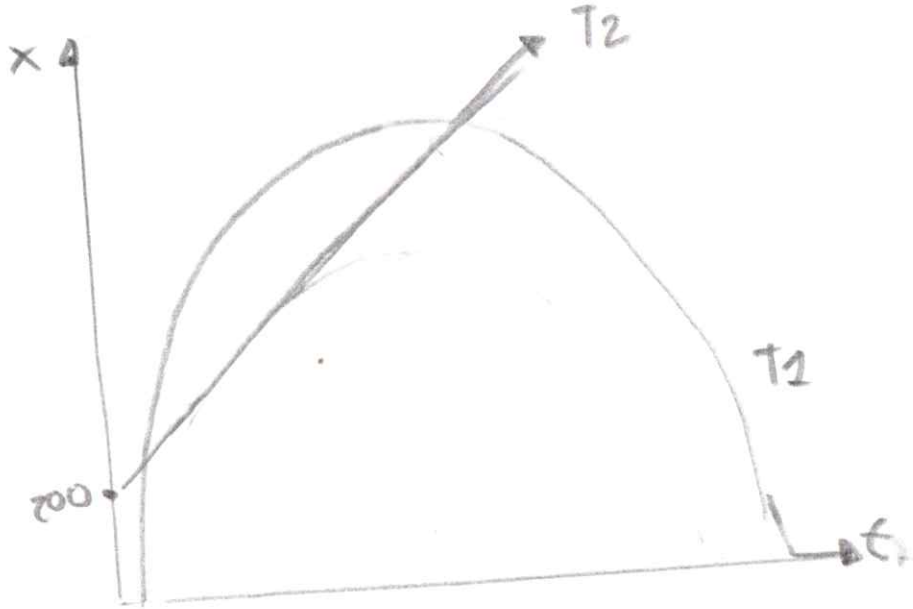
$x = x_2 = 15(22.54) + 200$

$x = 538.1m$

a) si las vacas testiguan un choque

b) a 538.1m fue el encuentro de antes pegarle al tren de carga

c)



6)

$d = v_0 t + \frac{1}{2} a t^2 + d_0$

$v_f = v_0 + a t$

$v_f^2 = v_0^2 + 2 a \Delta d$

$a = -0.100 \frac{m}{s^2}$

tren de pasajeros: $x = 25t + \frac{1}{2}(-0.100)t^2 + 0$

$x_1 = 25t - 0.5$ $\quad 25t + (-0.100)t$

$v_{f1} = 25 - 0.100t$

$v_{f1}^2 = 625 - 2(0.100)(x_2 - x_1)$

$v_{f1}^2 = 625 - 0.200x$

tren de carga: $x = v_0 t + x_0$ $\quad x_2 = 15t + 200$

$x_1 = x_2$

$25t - 0.5t^2 = x_2 \quad 15t + 200 \rightarrow 0.05t^2 - 10t + 200 = 0$